

A LECTURE FROM COAST

The Polya Method

a quiet pattern, restored to its full sequence



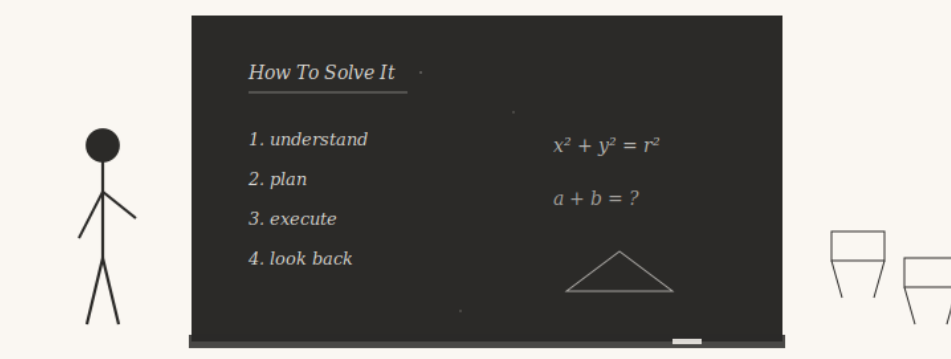
*Mathematicians do not leap. They follow a pattern so quiet that
for two thousand years no one wrote it down — until a
Hungarian refugee at Stanford did.*

A Hungarian mathematician at Stanford

George Polya was born in Budapest in 1887. He was a Hungarian Jew, which meant that in 1940, with Europe burning, he and his wife sailed to America. He was fifty-three years old and arguably the most respected combinatorialist alive. Stanford gave him a chair, and he started teaching American undergraduates.

He noticed something he had not expected.

His students — brilliant, well-prepared, ambitious — got stuck on the same kinds of problems, in the same kinds of ways, over and over. When he watched them work, he saw what was happening. They would read a problem once, lunge at it with the first technique that came to mind, fail, get frustrated, and give up. They never went back to the problem itself. They never checked their answer. They never asked whether their plan was even working.



How To Solve It, on the chalkboard. The method itself is shorter than the title.

When Polya watched real research mathematicians — including himself — he saw something different. Mathematicians moved slowly. They read the problem six times. They restated it in their own words. They drew pictures. They tried small cases. They named variables. They paused. They came back. They worked, then *looked back at the work*, every time.

It was a pattern. And no one had written it down.

Polya wrote it down. The book was 250 pages, came out in 1945, was called *How to Solve It*, and quietly became one of the most influential teaching texts of the twentieth century. Over a million copies have sold. It has never gone out of print. Every serious mathematician knows it. Almost no one outside mathematics has heard of it.

The pattern is four steps. Most people skip two of them. This lesson is the pattern, restored to its full sequence.

Module 2

A small demonstration

Before we go further, a problem:

“A bear walks 10 km south, then 10 km east, then 10 km north, and ends up exactly where it started. What colour is the bear?”

Don't try to solve it yet. Just sit with it for thirty seconds. Notice what your mind does.

Most people do one of two things. Either they guess (brown, because most bears are brown), or they get a vague feeling that this is a trick question and give up.

A mathematician — or anyone who has been through this lesson — does something completely different. They re-read it. They notice the word *exactly*. They draw it. They notice that the answer hinges on geography, not biology. They follow the geography until it produces the bear.

By the end of this lecture you will solve that problem, slowly, using a method that will also work on completely different problems — including problems that are not mathematical at all.



Module 3

Why your brain is built to skip

There is a reason no one teaches step 1 or step 4 of Polya's method, and the reason is not laziness — it is biology.

Your brain has a ‘do something’ reflex. Faced with a problem, it wants to pick the first plausible action and execute. This was useful when problems were lions in the grass. It is catastrophic when problems are abstract puzzles, code bugs, or career decisions, because the first plausible action is almost never the right one — and you spend hours chasing it instead of thinking.

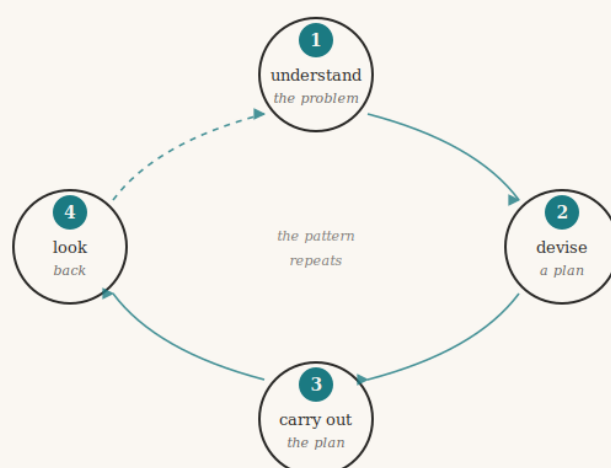
Polya's method is, fundamentally, a forcing function. It forces you to spend time on understanding before you act, and on reflection after you act. Those are the two stages your brain naturally rushes through. The method works because it makes you stay in those stages until you have actually been in them.

Almost everyone who claims to be ‘bad at problem-solving’ is not. They are skipping steps 1 and 4. We are going to put them back.

The four steps

Here is the method, in Polya's own ordering:

- 1. Understand the problem.** Read it. Restate it in your own words. Identify what is being asked, what is given, and what conditions hold. Draw a picture if you can. Notice the words that are precise.
- 2. Devise a plan.** Connect the problem to one you know. Look for patterns. Have you seen a related problem before? Could you solve a simpler version first? Could you work backwards from the answer?
- 3. Carry out the plan.** Execute. Check each step. Be honest about whether each step is justified or merely hopeful.
- 4. Look back.** Verify the answer. Does it make sense? Could you have arrived at it more simply? What did this problem teach you that will help with the next one?



The pattern is a cycle. The fourth step feeds the first one of the next problem.

The whole method fits on a postcard. Its power is not its content. Its power is the order, and the fact that you actually do all four — including the boring ones.

Module 5

Step 1, the slow read

We are going to apply step 1 to the bear problem.

Read the problem again, slowly:

“A bear walks 10 km south, then 10 km east, then 10 km north, and ends up exactly where it started. What colour is the bear?”

Now: restate it. What is being asked? *The colour of the bear.* What is given? *A specific walking path: 10 km south, 10 km east, 10 km north.* What is the key condition? *That the bear ends exactly where it started.*

That last word — *exactly* — is the whole problem. On a flat plane, the path described forms an open shape: you walk south, then east, then north back to the same latitude, but you are now 10 km east of where you started. You can't get back exactly. So the geometry of the surface must be doing the work.

Now draw it. Don't skip this. Pick a sheet of paper or imagine clearly: south 10, east 10, north 10. Where is the only place on Earth where this path closes?

Already, before step 2, the problem has narrowed dramatically. Most people who fail this problem fail because they never did step 1.

Module 6

Step 2, devise a plan

The clue is geography. The path closes. The bear is somewhere where you can walk south, then east, then north, and come back to your start.

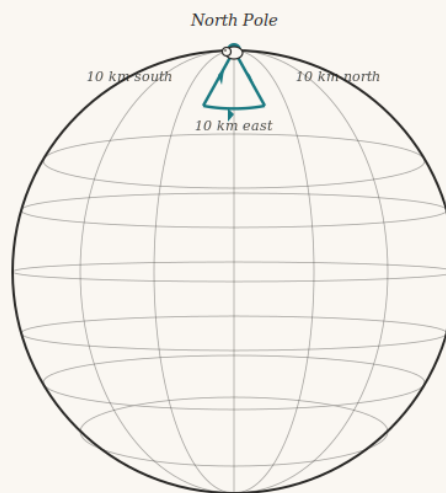
Have you seen a related problem? Yes — anyone who has held a globe has. Walk south from the North Pole 10 km, east 10 km (which curves around at high latitude), then north 10 km — and you are back at the North Pole.

Now we have a plan: assume the bear starts at the North Pole. Verify the geometry. Identify what bears live at the North Pole.

Module 7

Step 3, carry out the plan

At the North Pole, *south* is well-defined — every direction is south. Walk 10 km south. You are now on a small circle of latitude. Walk 10 km east along this latitude. You are still on the same circle, 10 km from your starting point along the parallel, but the parallel is a closed loop — so walking east is meaningful. Now walk 10 km north. Lines of longitude all converge at the North Pole, so 10 km north returns you to the pole.



Only at the North Pole does south-east-north close exactly.

The bear started at the North Pole. The only bears there are polar bears. The bear is white.

Module 8

Step 4, look back

This is the step most people never do. Don't skip it.

Look at what you did. Notice: the problem was not really about bears at all. It was a geometry problem disguised in zoology. The whole problem hinged on one word — *exactly*. Once you noticed that word, the rest was forced.

Notice also: the solution required no mathematics beyond what an eleven-year-old knows. The hard part was not the maths. The hard part was reading the problem properly, and the harder part was *believing* that reading it properly was where the work was.

Now ask: is there another solution? Could the bear be somewhere else? Surprisingly, yes — there are infinitely many starting points near the *South Pole* where the same path closes (rings where the eastward walk takes you exactly once, or twice, or n times around the pole). But no bears live near the South Pole. So our answer holds.

This is what looking back gets you. It deepens the problem. It also makes the next problem easier, because you now know that ‘weird trick questions’ often hide geometry, and that you should read every single word.

The method against a non-mathematical problem

Polya wrote the method for mathematics, but it works on anything.

Take a real problem: *I want to change careers, but I don't know what to switch to.*

Step 1 — understand the problem. What is actually being asked? Not ‘what should I do’ but more precisely: *what set of careers am I willing to consider, what do I want from a career, and what are my constraints?* Most people skip this and jump to a career list.

Step 2 — devise a plan. Have I seen a related problem? Yes: I've made career choices before. What worked? What didn't? Could I solve a simpler version — say, *what would I do if I had one year of guaranteed income?* Could I work backwards from a future self?

Step 3 — carry out the plan. Actually do the exercises from step 2. Write the constraints down. Talk to three people in three possible fields. Run the one-year experiment in imagination.

Step 4 — look back. After three weeks of investigation, was the problem the right one? Often, the act of doing step 3 reveals that the real problem was something else — like fear of failure, not lack of options. You won't know that without step 4.

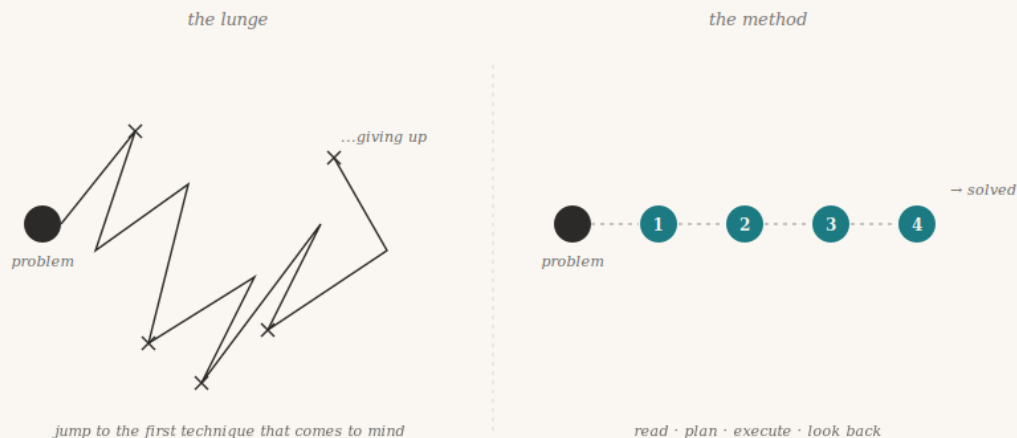


*The same four steps. A different domain.
The technique works on anything where your reflex would be to lunge.*

The four steps are not for maths. They are for any situation where your reflex would be to lunge.

A closing thought

There is something quietly heroic about Polya's method, and it is that it does not require you to be a genius. It requires you to be patient. To read carefully. To plan honestly. To execute deliberately. To look back. None of those are gifts. All of them are practices.



Most people lunge. The method is the alternative.

The students Polya watched at Stanford in the 1940s were extraordinary — they got into Stanford in an era when that meant something quite different — and they still got stuck on the same problems, in the same way, because no one had taught them to slow down. He spent the rest of his career fixing that, and the book he wrote became the silent companion of mathematicians for generations.

Most lessons promise to give you something. This one is unusual in that it mostly takes something away — the urgency to act before you understand. You will be slower. You will solve more problems.

Welcome to Polya's pattern.



What to learn next on Coast

- ◆ *First-Principles Thinking* — reasoning from the bedrock, not the analogy.
- ◆ *Fermi Estimation* — how to estimate any quantity in the world to within an order of magnitude.
- ◆ *The Feynman Technique* — exposing the gaps in your own understanding.